Quiz 2

Name: _____

Show your work. Unsupported answers may receive limited credit. **10 points + 3 extra credit**

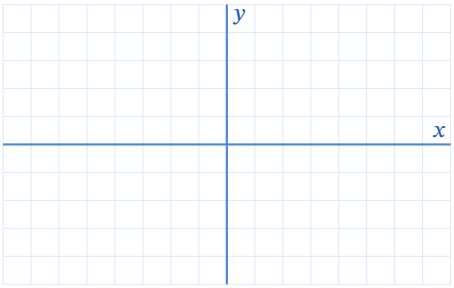
1. Piecewise function. **[5 points]**

Let

$$p(x) = \begin{cases} 2x + 1, & x < 0, \\ x^2 - 1, & x \geq 0. \end{cases}$$

(a) Find $p(-3)$, $p(0)$, and $p(2)$. Show which formula you use for each input. [2 points]

(b) Graph p. Use the correct open or closed point at the boundary. [1 point]



(c) State the domain, range, x-intercept(s), and y-intercept. [2 points]

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2. Constant rate of change. **[5 points]**

The values of a function f are shown below.

x	-1	0	1	2
$f(x)$	2	5	8	11

(a) Determine whether f is linear. Support your answer using rates of change. [2 points]

(b) If f is linear, find its slope. [1 point]

(c) Find a formula for $f(x)$. [1 point]

(d) State the y-intercept as an ordered pair. [1 point]

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Extra Credit. Information from a formula. **[3 points]**

Let

$$h(x) = \frac{x^2 - x - 6}{x + 1}.$$

(a) State the domain in interval notation. [2 points]

(b) Find the y-intercept. Write it as an ordered pair. [1 point]